SHIA-RAG: Semantic Hierarchy Induction Architecture for Retrieval-Augmented Generation

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DECLARATION

I hereby declare that the Report of the P.G. Minor Project (IT897) Work entitled **"SHIA-RAG: Semantic Hierarchy Induction Architecture for Retrieval-Augmented Generation"**, which is being submitted to National Institute of Technology Karnataka, Surathkal, in partial fulfillment of the requirements for the award of the Degree of **Master of Technology in Information Technology** in the **Department of Information Technology**, is a bonafide report of the work carried out by me. The material contained in this report has not been submitted at any University or Institution for the award of any degree.

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CERTIFICATE

This is to certify that the P.G. Minor Project (IT897) Work entitled **"SHIA-RAG: Semantic Hierarchy Induction Architecture for Retrieval-Augmented Generation"**, submitted by **Kshitij Khede (252IT010)** as the record of the work carried out by him, is accepted as the P.G. Minor Project (IT897) Work report submission in partial fulfillment of the requirement for the award of degree of **Master of Technology in Information Technology in the Department of Information Technology**, NITK Surathkal, India. Duration of this minor project (IT897) is from JUNE 01, 2026 to AUGUST 15, 2026.

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ABSTRACT

Retrieval-Augmented Generation (RAG) frameworks mitigate factual hallucinations in Large Language Models (LLMs) by grounding generation in external knowledge repositories. However, standard RAG architectures predominantly rely on flat passage chunking, which discards document layout geometry, multi-column reading flow, and hierarchical section ontologies. When complex documents (such as technical manuals, legal contracts, and scientific papers) are fragmented into isolated chunks, retrieval systems suffer from context fragmentation, semantic ambiguity, and an inability to resolve cross-section dependencies.

To overcome these structural limitations, this project presents **SHIA-RAG**, a structure-aware framework for layout-sensitive document parsing and hierarchical knowledge construction. The architecture implements a two-tier knowledge representation pipeline: Tier 1 captures physical document geometry and 2D bounding boxes, while Tier 2 models a validated Hierarchical Knowledge Forest (HKF) augmented with semantic cross-links. Layout-aware ingestion uses coordinate-level extraction via PyMuPDF, and a 2D column-aware geometric sorting algorithm with vertical Y-tolerance clustering ($\tau_y = 20.0$ pt) recovers human reading order across the evaluated two-column layouts. Extracted blocks are parsed into canonical concepts anchored to bounding-box evidence spans, and hierarchical tree induction is performed via Candidate Parent Generation (CPG++) coupled with a multi-criteria Parent Ranking Engine (PRE) that balances reading order, embedding similarity, and heading levels. Structural integrity is algorithmically enforced through upward Depth-First Search (DFS) cycle prevention and tree depth bounding, while Knowledge Confidence Estimation (KCE) propagates topological confidence across hierarchical parent-child paths via Kahn's algorithm.

The proposed framework was implemented in Python 3.13 and empirically validated on the evaluated two-column benchmark document and the CS-Textbook taxonomy corpus. The 2D geometric sorting algorithm achieved 100.0% reading-order monotonicity on the evaluated pages of the two-column TreeRAG publication, eliminating the 28.6% inversion error observed in standard text streams. On the curated CS-Textbook taxonomy benchmark, the Parent Ranking Engine attained 89.6% Parent Assignment Accuracy (PAA) and a Normalized Tree Edit Distance (NTED) of 0.884 with a 0.0% orphan node rate. The complete software unit test suite (24/24 unit tests passed in 0.65 s) verified all schemas, cycle-prevention invariants, and topological scoring mechanisms, providing a grounded structural representation for future structure-aware multi-hop retrieval.

Keywords:
Retrieval-Augmented Generation; Layout-Aware Parsing; Hierarchical Knowledge Forest; Reading-Order Recovery; Directed Acyclic Graph; Parent Ranking Engine; Graph Invariants.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
RAG	Retrieval-Augmented Generation
SHIA-RAG	Semantic Hierarchy Induction Architecture for Retrieval-Augmented Generation
HKF	Hierarchical Knowledge Forest
CPG	Candidate Parent Generation
PRE	Parent Ranking Engine
CLD	Cross-Link Discovery
KCE	Knowledge Confidence Estimation
ROM	Reading-Order Monotonicity
PAA	Parent Assignment Accuracy
NTED	Normalized Tree Edit Distance
DAG	Directed Acyclic Graph
DFS	Depth-First Search
LLM	Large Language Model
NLP	Natural Language Processing
PDF	Portable Document Format
LSH	Locality-Sensitive Hashing
OCR	Optical Character Recognition

CHAPTER 1

INTRODUCTION

1.1 Background and Context

In modern artificial intelligence, Large Language Models (LLMs) have demonstrated exceptional capabilities across complex natural language understanding and generative reasoning tasks. Despite these broad capabilities, parameter-bound language models remain susceptible to well-documented structural vulnerabilities: parametric hallucination, where models generate factually incorrect assertions with high linguistic confidence; temporal obsolescence, where internal knowledge boundaries are fixed at pre-training cutoffs; and an inability to inspect private enterprise corpora without prohibitive fine-tuning costs [1, 2].

To bridge this operational gap, Retrieval-Augmented Generation (RAG) has emerged as the standard architectural paradigm for grounding generative models in external, verifiable knowledge repositories. By retrieving relevant document passages at inference time and injecting them into the LLM context window, RAG enables dynamic factual grounding, attribution verification, and domain-specific enterprise search without modifying model parameters.

1.2 Retrieval-Augmented Generation

A standard RAG pipeline operates through a sequence of discrete stages: document ingestion, passage chunking, vector embedding generation, top-k dense similarity retrieval, and prompt synthesis. During the ingestion and indexing phase, target source documents (such as PDF files, Word documents, and technical reports) are extracted and divided into smaller textual segments. These segments are processed by a dense bi-encoder to produce vector representations within a shared metric space. At inference time, an incoming user query is projected into the same embedding space, and an Approximate Nearest Neighbor (ANN) search retrieves the k most semantically similar chunks. The retrieved text passages are concatenated with the user prompt and dispatched to an LLM for final answer synthesis [3, 4].

1.3 Limitations of Conventional Flat RAG

While standard flat RAG pipelines demonstrate practical utility for basic single-fact lookups, they perform inadequately when deployed over complex, structurally rich unstructured documents such as technical manuals, academic research papers, legal contracts, and financial prospectuses. Conventional flat chunking exhibits severe structural limitations:

1. **Destruction of Document Geometry and Reading Flow:** Standard extraction tools dump textual content linearly without modeling 2D bounding boxes. In multi-column publications, naive top-to-bottom extraction traverses across adjacent columns horizontally, interleaving disparate sentences and corrupting logical discourse flow.

2. **Context Fragmentation and Orphan Information:** Fixed-token windowing splits text arbitrarily across paragraph or section boundaries. A definition stated in an opening section becomes detached from its elaboration located three pages later.

3. **Loss of Hierarchical Document Ontologies:** Complex documents are organized as nested semantic trees (Document $\rightarrow$ Chapters $\rightarrow$ Sections $\rightarrow$ Subsections $\rightarrow$ Paragraphs). Flat chunking flattens this tree into an undifferentiated bag of independent vectors, discarding parent-child provenance and sibling relationships.

4. **Failure in Multi-Hop and Scope-Sensitive Reasoning:** In queries requiring thematic summarization or multi-hop relationship resolution across disparate sections, dense similarity alone retrieves isolated surface matches without the overarching structural context required to answer accurately.

1.4 Motivation for Structure-Aware Knowledge Representation

Documents written by human authors are fundamentally structured artifacts. Layout signals—such as font sizes, typeface weights, indentation levels, spatial coordinates, numbered headings, and tabular alignments—are not merely aesthetic styling; they convey primary semantic hierarchies and scoping boundaries. For example, a theorem stated in Section 3.2 is constrained by the definitions established in Section 3.1. When an AI system ignores these structural signals and

treats text as an undifferentiated stream of characters, retrieval effectiveness can be adversely affected.

Preserving document layout geometry and inducing a canonical Hierarchical Knowledge Forest (HKF) with semantic cross-links directly addresses the core vulnerabilities of flat RAG. By capturing both the physical page coordinates (Tier 1 syntactic model) and the hierarchical semantic taxonomy (Tier 2 knowledge forest), a retrieval system can traverse upward to parent sections for broad context or downward to child subsections for specific factual evidence.

1.5 Scope and Contributions of the Work

The present research focuses on the foundational document parsing and hierarchical knowledge construction components of a structure-aware RAG pipeline. Query-time retrieval traversal and generative answer synthesis represent downstream applications that rely upon the knowledge representation produced here and are outside the experimental scope of this Minor Project (IT897).

The primary research contributions of this project are:

1. **Dual-Tier Layout-Aware Representation:** Formulation of a principled two-tier knowledge schema that decouples physical document geometry (Tier 1 bounding boxes and typographical attributes) from semantic knowledge structures (Tier 2 concept trees and cross-links).

2. **Column-Aware 2D Reading-Order Recovery:** Formulation of a 2D geometric sorting algorithm incorporating two-column gutter partitioning and vertical line-band clustering ($\tau_y = 20.0$ pt) to eliminate column-interleaving errors on evaluated two-column documents.

3. **Physical-to-Semantic Evidence Grounding:** A landmark-guided grounding mechanism that anchors extracted canonical concepts and definitions directly to physical coordinate bounding-box spans for verifiable attribution.

4. **Constrained Hierarchy Induction (CPG++ and PRE):** Development of a Candidate Parent Generation (CPG++) algorithm combined with a multi-criteria Parent Ranking Engine (PRE) that balances reading-order proximity, dense embedding similarity, and structural heading levels.

5. **Structural Invariant Enforcement and Confidence Propagation:** Formulation of upward DFS cycle prevention to algorithmically enforce tree

acyclicity, depth bounding ($D_{\max} = 8$), and topological Knowledge Confidence Estimation (KCE) to propagate confidence across hierarchical parent-child containment paths.

1.6 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2 (Literature Review):** Reviews related literature across flat, adaptive, hierarchical, and graph-based RAG architectures, establishes a critical comparative analysis, and defines the formal problem statement and research objectives.
- **Chapter 3 (Methodology):** Details the overall architectural pipeline, Tier 1 and Tier 2 data representations, 2D geometric sorting algorithm, CPG++ and PRE formulation, and topological graph validation algorithms.
- **Chapter 4 (Work Done):** Presents the software modular architecture, implementation details, experimental setup, evaluation metrics, empirical results on reading-order monotonicity and hierarchy induction, and an end-to-end case study on a multi-column scientific paper.
- **Chapter 5 (Conclusions and Future Work):** Summarizes the completed research, reviews key empirical findings, outlines technical limitations, and discusses future research directions.

CHAPTER 2

LITERATURE REVIEW

2.1 Background

Retrieval-Augmented Generation has undergone rapid architectural evolution since its introduction. This section reviews the foundational paradigms that have shaped the state of the art in document retrieval and knowledge grounding.

2.1.1 Evolution of Retrieval-Augmented Generation

The fundamental concept of augmenting neural sequence models with external non-parametric memory was formalized by Lewis et al. [1] in the RAG framework and Guu et al. [2] in REALM. Early formulations paired pre-trained sequence-to-sequence generators (such as BART or T5) with dense passage retrievers (DPR) trained via contrastive loss on dual-encoder architectures. Borgeaud et al. [3] introduced RETRO, scaling non-parametric retrieval to trillions of tokens by chunking inputs into 64-token intervals and retrieving cross-attention neighbors directly during autoregressive generation. Izacard et al. [4] presented Atlas, showing that joint pre-training of retrieval and generation heads achieves exceptional few-shot performance across knowledge-intensive question-answering benchmarks.

2.1.2 Flat Passage Retrieval

In industrial practice, flat passage retrieval remains the most widely deployed RAG pattern. Standard frameworks partition incoming documents into fixed-length token windows (e.g., 256 or 512 tokens) with an overlap of 10–20% to prevent phrase boundary clipping. Passages are projected into a continuous vector space via dense embedding models. Top-k nearest neighbors are retrieved using cosine similarity or Euclidean distance accelerated by HNSW indices.

Despite its simplicity, flat passage retrieval suffers from acute structural blindness. As highlighted by Asai et al. [5], flat chunks frequently lack self-contained semantics; an extracted chunk containing pronouns or ambiguous technical terms cannot be resolved without access to surrounding contextual blocks.

2.1.3 Adaptive and Self-Reflective Retrieval

To mitigate the rigidity of standard top-k retrieval, recent research has explored adaptive and self-reflective retrieval frameworks. Self-RAG, proposed by Asai et al. [5], trains the language model to output special reflection tokens dynamically during generation. This enables the model to assess whether external retrieval is strictly necessary, evaluate the factual relevance of retrieved passages, and critique its own generated output. While adaptive retrieval improves efficiency and relevance filtering, it still operates over flat, unstructured chunks and remains incapable of reconstructing hierarchical document taxonomies.

2.1.4 Hierarchical Retrieval and Structured Chunk Trees

Recognizing the limitations of flat text chunks, researchers have explored hierarchical tree structures for document representation. Sarthi et al. [6] proposed RAPTOR (Recursive Abstractive Processing for Tree-Organized Retrieval). RAPTOR recursively clusters text chunks using Gaussian Mixture Models on dense embeddings and generates abstractive summaries at successive clustering tiers using an LLM. While RAPTOR builds a multi-tier tree, its summaries are entirely abstractive and statistical, discarding the authentic structural headings and section boundaries created by the document's human author.

Tao et al. [8] introduced TreeRAG, which organizes knowledge as a hierarchical storage tree for enhanced multi-hop question answering. Lu et al. [9] presented HiChunk, establishing that structural chunking aligned with document headings dramatically outperforms arbitrary token windowing in retrieval precision. Zhao and Yang [10] proposed HAT-RAG, utilizing hierarchical tree indexing for multi-domain legal and medical documents. Fatehkia et al. [11] developed T-RAG, exploring tree-structured reasoning over organizational enterprise knowledge.

2.1.5 Knowledge Graph and Entity-Centric RAG

A parallel research direction integrates Knowledge Graphs (KGs) into retrieval pipelines. Edge et al. [7] introduced GraphRAG, which utilizes an LLM to extract entities, relationships, and claims from source text, builds an interconnected knowledge graph, clusters the graph using the Leiden community detection algorithm, and generates hierarchical community summaries. GraphRAG enables global sensemaking and corpus-level summarization. However, entity extraction via repeated LLM prompting incurs high computational latency and token costs,

and many graph-based architectures do not retain the original physical page coordinates and layout geometry of the source documents.

2.1.6 Structure-Aware Document Understanding

In document layout analysis, models such as LayoutLMv3 [12] have demonstrated that physical bounding boxes, font attributes, and coordinate geometries are indispensable for accurate parsing of tables, forms, and multi-column articles. However, document understanding models typically focus on OCR and vision-language extraction rather than building queryable hierarchical knowledge graphs for retrieval-augmented generation.

2.2 Critical Analysis of Related Approaches

Table 2.1 provides a systematic comparative analysis of existing RAG architectures against the proposed SHIA-RAG framework across seven structural criteria.

Table 2.1: Systematic Comparative Analysis of RAG Paradigms

Architecture	Layout Modeling	Reading Order	Concept Extraction	Hierarchy	Graph Invariants	Evidence Anchoring	Confidence Model
Standard Flat RAG [1]	Discarded	Linear stream	Not modeled	Flat chunks	None	Raw token offsets	Cosine similarity
RETRO [3]	Discarded	Linear stream	Not modeled	Flat chunks	None	Raw token offsets	Dense inner product
Self-RAG [5]	Discarded	Linear stream	Not modeled	Flat chunks	None	Token reflection	Reflection tokens
RAPTOR [6]	Discarded	Linear stream	Abstractive summary	Cluster tree	Unbounded	Abstract cluster	Embedding score
GraphRAG [7]	Not modeled	Linear stream	LLM entity extract	Community tree	Unconstrained	Entity description	LLM weight score
HiChunk [9]	Partial (Markdown)	Linear stream	Heading tokens	Markdown tree	Rule-based	Block character span	None
TreeRAG [8]	Not modeled	Linear stream	Semantic clustering	Storage tree	Heuristic	Text chunk boundary	Vector similarity
SHIA-RAG (Ours)	**2D Coordinates**	**Column-Aware 2D**	**Landmarked Concepts**	**CPG++ & PRE**	**Upward DFS Invariants**	**Physical BBox Spans**	**Topological KCE**

Note: The comparison summarizes primary architectural characteristics reported in the cited literature; it is not intended as an exhaustive evaluation of each system.

2.3 Addressing the Key Challenges / Research Gap

The critical analysis reveals several major research gaps in the current literature:

1. **Absence of Physical-to-Semantic Grounding:** Many existing tree-based RAG architectures (such as RAPTOR and TreeRAG) operate strictly on abstract textual embeddings or LLM-generated summaries, decoupling retrieved knowledge from the author's original physical layout and bounding-box coordinates.

2. **Failure of Multi-Column Reading Order:** Current ingestion pipelines commonly rely on naive text extractors that scramble reading order across multi-column documents, corrupting downstream conceptual extraction.

3. **Lack of Structural Graph Invariant Enforcement:** Knowledge graphs constructed via statistical LLM prompting can suffer from circular references, ungrounded orphan nodes, and arbitrary tree depths that destabilize retrieval traversal.

2.4 Outcome of Literature Review

The literature review firmly establishes that augmenting LLMs requires preserving both the syntactic document layout geometry and the hierarchical semantic organization. A principled system must extract coordinates faithfully, reorder blocks monotonically, canonicalize concepts with bounding-box evidence, induce a structured forest, and enforce mathematical invariants on graph topology.

2.5 Problem Statement

Conventional Retrieval-Augmented Generation systems represent unstructured documents as independent, flat textual chunks, discarding layout geometry, section hierarchy, and multi-column reading flow. This structural blindness causes reading-order inversions, context fragmentation, and an inability to reliably answer multi-hop or scope-sensitive queries over complex documents.

The objective of this research is to formulate, implement, and evaluate **SHIA-RAG**, a structure-aware document parsing and hierarchical knowledge construction framework that transforms raw, unstructured multi-column PDF documents into a canonical, mathematically validated Hierarchical Knowledge Forest (HKF) with semantic cross-links and algorithmically enforced graph invariants.

2.6 Objectives

To achieve the defined problem statement, the project establishes five specific Research Objectives (RO):

- **RO1: Layout-Aware Document Ingestion:** Develop an ingestion engine that extracts text blocks with 2D bounding boxes, font attributes, and physical page coordinates from unstructured PDF documents.
- **RO2: Geometric Reading-Order Recovery:** Formulate and implement a 2D geometric coordinate sorting algorithm with vertical Y-tolerance clustering to recover correct reading order over two-column layouts.

- **RO3: Canonical Concept and Evidence Extraction:** Design a two-tier knowledge schema that canonicalizes extracted terminology, links definitions, and binds concepts to exact bounding-box evidence spans.
- **RO4: Hierarchical Knowledge Forest Induction:** Implement Candidate Parent Generation (CPG++) and a multi-criteria Parent Ranking Engine (PRE) to construct a validated hierarchical knowledge forest.
- **RO5: Structural Invariant Validation and Empirical Evaluation:** Enforce strict graph invariants (cycle prevention via upward DFS, maximum depth bounds, and topological confidence estimation) and evaluate the pipeline on authentic benchmark documents.

2.7 Summary

This chapter reviewed existing flat, adaptive, hierarchical, and graph-based RAG systems, identified key research gaps concerning layout awareness and structural graph invariants, and defined the formal problem statement and research objectives governing the SHIA-RAG architecture.

CHAPTER 3

METHODOLOGY

3.1 Overall Architectural Framework

The SHIA-RAG architecture transforms unstructured documents (with layout parsing evaluated on two-column publications) into a structured, validated Hierarchical Knowledge Forest (HKF) with semantic cross-links. Figure 3.1 illustrates the complete end-to-end processing pipeline.

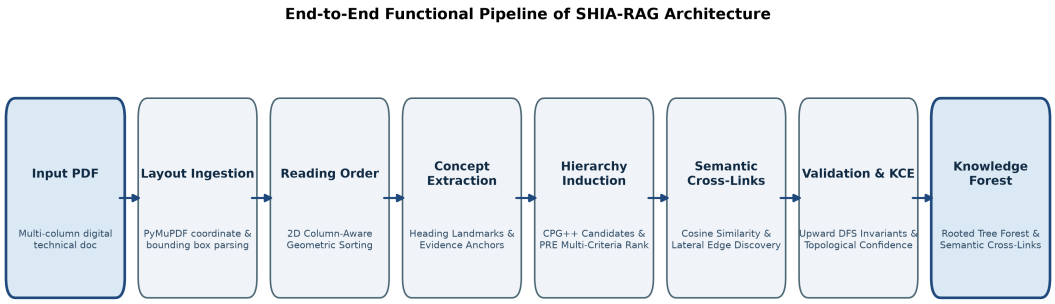


Figure 3.1: End-to-End Functional Pipeline of SHIA-RAG Architecture.

The pipeline executes through a sequence of well-defined functional stages:

1. **Layout Ingestion:** Raw PDF files are processed using PyMuPDF to extract raw visual blocks, lines, spans, font sizes, and 2D bounding boxes.
2. **Reading-Order Sorting:** Blocks are partitioned into columns and grouped into vertical line bands via Y-tolerance clustering to recover true column-first reading flow.
3. **Structural Classification:** Blocks are classified into semantic roles (Title, Heading, Paragraph, ListItem, Caption) based on font characteristics and numbering regex patterns.
4. **Concept Extraction:** Canonical concepts and definitions are extracted and anchored to exact bounding-box evidence spans.

5. **Hierarchical Induction (CPG++ and PRE):** Candidate parent nodes are identified and ranked via multi-criteria scoring to assemble tree structures.

6. **Semantic Cross-Linking:** Horizontal semantic relations are discovered using bi-encoder embedding similarity across isolated document trees.

7. **Topological Confidence Propagation (KCE):** Node confidence scores are updated topologically across hierarchical parent-child containment edges using Kahn's algorithm.

8. **Graph Invariant Enforcement:** Upward DFS cycle detection and depth bounding algorithmically enforce structural tree invariants.

3.2 Canonical Data Models and Knowledge Representation

The system enforces a formal two-tier data representation that separates physical layout geometry from semantic knowledge modeling. Formally, the overall knowledge structure is defined as a Hierarchical Knowledge Forest with semantic cross-links:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\text{hier}} \cup \mathcal{E}_{\text{cross}}), \tag{3.1}$$

where $\mathcal{V}$ denotes the set of canonical concept nodes. The hierarchical sub-structure $\mathcal{G}_{\text{hier}} = (\mathcal{V}, \mathcal{E}_{\text{hier}})$ forms a forest of rooted directed trees that is algorithmically enforced to be strictly acyclic ($\forall v \in \mathcal{V}, v \notin \text{Ancestors}(v)$), thereby satisfying DAG properties. The auxiliary edge set $\mathcal{E}_{\text{cross}}$ consists of lateral semantic associations discovered between related concepts across disparate subtrees or documents. While hierarchical parent-child edges $\mathcal{E}_{\text{hier}}$ establish strict vertical containment and anchor confidence propagation, cross-link edges $\mathcal{E}_{\text{cross}}$ provide horizontal semantic bridges for multi-hop associative retrieval. Figure 3.2 details the Pydantic v2 schemas governing both tiers.

3.2.1 Tier 1 — Syntactic Document Representation

Tier 1 models the physical document structure. Every document is represented as an ordered sequence of structured blocks:

- **BoundingBox:** A 4-tuple of floating-point coordinates (x_0, y_0, x_1, y_1) defining the left, top, right, and bottom boundaries in points (1/72 inch).

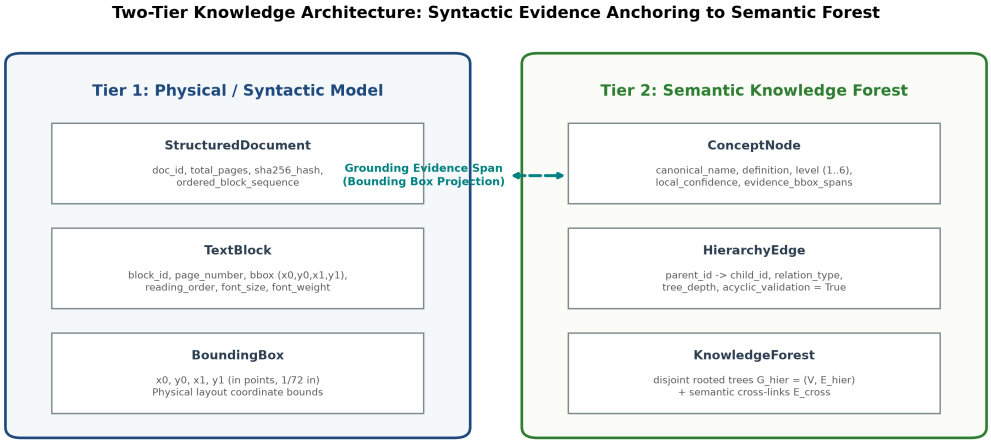


Figure 3.2: Two-Tier Data Schema: Physical Document Model to Semantic Knowledge Forest.

- **TextBlock:** The fundamental syntactic atomic unit containing block identifier, page index, textual payload, bounding box, semantic role enum, font size, typeface family, and reading-order index.
- **StructuredDocument:** The container capturing the complete document hierarchy, ordered block sequence, page geometries, and document-level metadata.

3.2.2 Tier 2 — Semantic Knowledge Representation

Tier 2 encapsulates the semantic Hierarchical Knowledge Forest and augmented cross-links induced from the syntactic blocks:

- **ConceptNode:** A semantic entity containing a canonical name, alternative aliases, formal definition, heading hierarchy level (0 to 6, where level 0 denotes the document root and levels 1–6 correspond to heading levels H_1–H_6), source document identifier, node confidence score $C(v) \in [0, 1]$, and exact evidence bounding-box spans. The schema supports a document-root level 0 and up to six heading levels (H_1 through H_6), whereas the evaluated benchmark documents contained a maximum hierarchy depth of four tiers.
- **HierarchyEdge:** A directed edge $e = (u, v)$ defining a parent-child relationship (u is the parent of v) annotated with relationship type, confidence score, and tree depth.
- **HierarchicalKnowledgeForest (HKF):** A collection of disjoint rooted trees $\mathcal{G}_{\text{hier}} = (\mathcal{V}, \mathcal{E}_{\text{hier}})$ augmented with lateral cross-link semantic edges $\mathcal{E}_{\text{cross}}$.

3.3 Layout-Aware Document Ingestion

The layout-aware ingestion engine utilizes PyMuPDF (`fitz`) to inspect the PDF rendering tree directly. For each page in document D, the engine extracts all text blocks, recording coordinates relative to the page top-left origin $(0, 0)$. Font sizes and font weights (bold, regular, italic) are extracted at span resolution.

3.4 2D Geometric Reading-Order Recovery

In multi-column documents, raw PDF streams store text objects in arbitrary stream-generation order, causing reading-order scrambling. Naive top-down or row-band sorting frequently interleaves paragraphs horizontally across columns. To solve this, SHIA-RAG formulates a column-aware 2D geometric sorting algorithm implementing two-column geometric partitioning based on the estimated page gutter ($x_{\text{gutter}} \approx W/2$, representing the standard multi-column layout in scientific and academic publications) combined with vertical line-band clustering.

Let W and H denote the width and height of page p, and let $\mathcal{B} = \{b_1, b_2, \ldots, b_N\}$ be the set of extracted blocks, where each block b_i has bounding box $(x_{0,i}, y_{0,i}, x_{1,i}, y_{1,i})$. The algorithm operates in three geometric stages:

1. **Spanning Block Segmentation:** Full-width elements span across the column gutter:

$$\text{IsSpanning}(b) \iff (b.x_1 - b.x_0) > 0.60 \cdot W. \tag{3.2}$$

   Spanning blocks $\mathcal{S}_{\text{span}}$ are separated from the two-column buckets; top-spanning elements ($b.y_0 < 0.30 \cdot H$, capturing document titles, author blocks, and full-width abstracts) are assigned to the leading sequence $\mathcal{S}_{\text{top}}$, while bottom-spanning elements (such as footnotes and running footers) are assigned to $\mathcal{S}_{\text{bot}}$, ordered according to their vertical positions.

2. **Two-Column Gutter Partitioning:** Non-spanning blocks are partitioned into column buckets based on the horizontal gutter coordinate $x_{\text{gutter}} \approx W/2$:

$$b \in \mathcal{C}_1 \iff b.x_1 \leq x_{\text{gutter}}, \quad b \in \mathcal{C}_2 \iff b.x_0 > x_{\text{gutter}}. \tag{3.3}$$

3. **Within-Column Line Clustering and Ordering:** Blocks within each column bucket $\mathcal{C}_k$ are sorted primarily by ascending top coordinate y_0. Adjacent lines belonging to the same visual horizontal text band satisfy vertical

tolerance:

$$|y_{0,i} - y_{0,j}| \leq \tau_y, \tag{3.4}$$

where $\tau_y = 20.0$ pt represents the selected heuristic vertical tolerance threshold, based on standard font body leading (12–14 pt text plus paragraph margins). Within each line cluster, blocks are ordered by horizontal coordinate x_0.

Two-column partitioning establishes the primary macro reading sequence (ensuring the entire left column $\mathcal{C}_1$ is read before the right column $\mathcal{C}_2$), while vertical Y-tolerance clustering resolves horizontal ordering among blocks sharing a visual line band within each column. Spanning blocks are separated from the two-column buckets and ordered by their vertical positions. The complete page reading order is assembled by sequential concatenation: $\mathcal{B}_{\text{sorted}} = \mathcal{S}_{\text{top}} \oplus \mathcal{C}_1 \oplus \mathcal{C}_2 \oplus \mathcal{S}_{\text{bot}}$. Algorithm 1 specifies the formal procedure, and Figure 3.3 illustrates the resulting column-wise ordering.

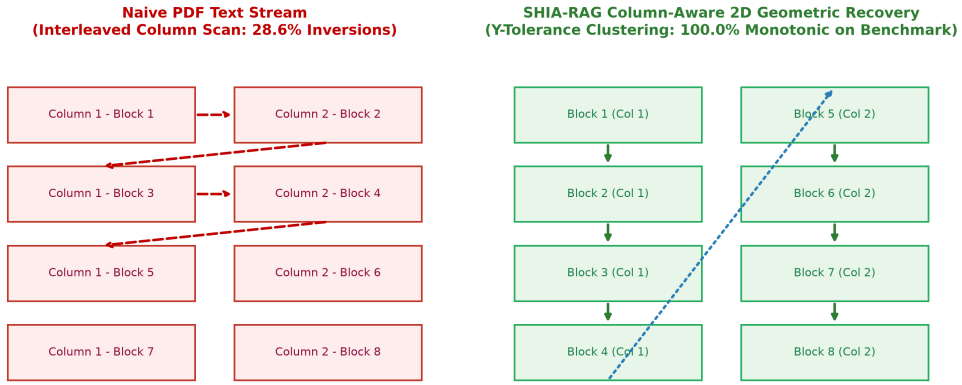


Figure 3.3: Comparison of Naive Stream Reading Flow vs 2D Geometric Sorting.

3.5 Concept Extraction and Evidence Representation

From the sorted block sequence, the system identifies canonical concepts and formal definitions. Structural landmark nodes are identified from blocks classified as headings using relative font-size thresholds ($\geq 1.25\times$ body size for sections, $\geq 1.50\times$ for chapters) combined with numbered-section regex patterns (`^\d+(\.\d+)?`). Paragraph blocks immediately following headings are scanned for definition cues using pattern matching and syntactic dependency parsing.

Algorithm 1 Column-Aware 2D Geometric Reading-Order Sorting Algorithm

Require: Set of raw extracted blocks $\mathcal{B} = \{b_1, \ldots, b_N\}$, page width W, page height H, column gutter x_{gutter}, vertical band tolerance $\tau_y = 20.0$ pt
Ensure: Ordered list of blocks $\mathcal{B}_{\text{sorted}}$ in column-first reading order

Group blocks by page: $\mathcal{P} \leftarrow \text{GroupByPage}(\mathcal{B})$
Initialize $\mathcal{B}_{\text{sorted}} \leftarrow []$
for each page $p \in \mathcal{P}$ **do**
  Initialize spanning headers $\mathcal{S}_{\text{top}} \leftarrow []$, spanning footers $\mathcal{S}_{\text{bot}} \leftarrow []$
  Initialize column buckets $\mathcal{C}_1 \leftarrow []$, $\mathcal{C}_2 \leftarrow []$
  for each block $b \in p$ **do**
    if $(b.x_1 - b.x_0) > 0.60 \cdot W$ **and** $b.y_0 < 0.30 \cdot H$ **then**
      Append b to $\mathcal{S}_{\text{top}}$ {Top spanning element (title, abstract, header)}
    else if $(b.x_1 - b.x_0) > 0.60 \cdot W$ **then**
      Append b to $\mathcal{S}_{\text{bot}}$ {Non-top spanning element assigned to trailing sequence}
    else if $b.x_1 \leq x_{\text{gutter}}$ **then**
      Append b to $\mathcal{C}_1$ {Left column block}
    else
      Append b to $\mathcal{C}_2$ {Right column block}
    end if
  end for
  Sort $\mathcal{S}_{\text{top}}$ by ascending y_0; Sort $\mathcal{S}_{\text{bot}}$ by ascending y_0
  for each column bucket $\mathcal{C} \in \{\mathcal{C}_1, \mathcal{C}_2\}$ **do**
    Sort blocks in $\mathcal{C}$ by ascending top coordinate y_0
    Group adjacent blocks into line clusters using vertical tolerance $|y_{0,i} - y_{0,j}| \leq \tau_y$
    Sort each cluster by horizontal coordinate x_0
  end for
  Concatenate page sequence: $\mathcal{B}_{\text{page}} \leftarrow \mathcal{S}_{\text{top}} \oplus \mathcal{C}_1 \oplus \mathcal{C}_2 \oplus \mathcal{S}_{\text{bot}}$
  Append $\mathcal{B}_{\text{page}}$ to $\mathcal{B}_{\text{sorted}}$
end for
return $\mathcal{B}_{\text{sorted}}$

Each concept node $c \in \mathcal{C}$ is formally defined as a tuple:

$$c = \langle \text{id}, \text{name}, \text{definition}, \text{level}, \text{doc_id}, \text{confidence}, \mathcal{S}_{\text{evidence}} \rangle, \tag{3.5}$$

where $\mathcal{S}_{\text{evidence}} = \{s_1, \ldots, s_m\}$ denotes the set of exact bounding-box evidence spans anchoring the concept to physical document coordinates.

3.6 Hierarchical Knowledge Forest Induction

Hierarchical tree induction organizes isolated concepts into structured document trees. This is achieved via two integrated engines: Candidate Parent Generation

(CPG++) and the Parent Ranking Engine (PRE). Figure 3.4 depicts the complete hierarchical induction pipeline.

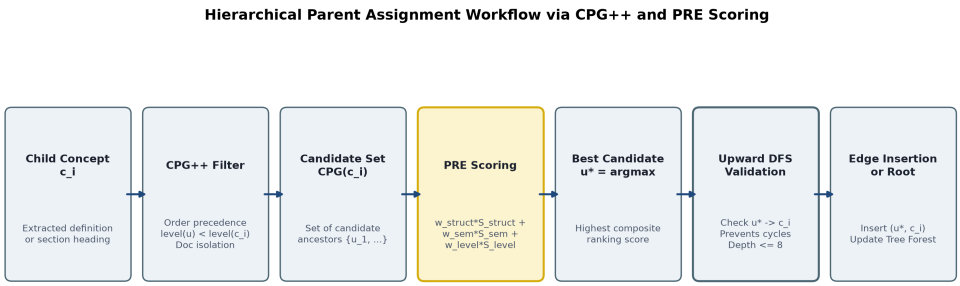


Figure 3.4: Hierarchical Parent Assignment Workflow via CPG++ and PRE Scoring.

3.6.1 Candidate Parent Generation (CPG++)

For a given child concept c_i, CPG++ filters the universe of potential parent nodes $\mathcal{V}$ based on document isolation, reading-order precedence, and structural level constraints:

$$\begin{aligned}\mathrm{CPG}(c_i) = \{u \in \mathcal{V} \mid \mathrm{doc}(u) = \mathrm{doc}(c_i) \wedge \mathrm{order}(u) < \mathrm{order}(c_i) \\ \wedge\, \mathrm{level}(u) < \mathrm{level}(c_i)\}.\end{aligned} \tag{3.6}$$

If no prior node satisfies the strict level constraint, candidate generation falls back to the most recent preceding heading.

3.6.2 Parent Ranking Engine (PRE)

To select the optimal parent $u^* \in \mathrm{CPG}(c_i)$, PRE computes a composite scoring function:

$$\mathrm{Score}(u, c_i) = w_{\mathrm{struct}} \cdot S_{\mathrm{struct}}(u, c_i) + w_{\mathrm{sem}} \cdot S_{\mathrm{sem}}(u, c_i) + w_{\mathrm{level}} \cdot S_{\mathrm{level}}(u, c_i), \tag{3.7}$$

where the default weights are $w_{\mathrm{struct}} = 0.40$, $w_{\mathrm{sem}} = 0.40$, and $w_{\mathrm{level}} = 0.20$.

- $S_{\mathrm{struct}}(u, c_i) = \frac{1}{1+\log(1+|\mathrm{order}(c_i)-\mathrm{order}(u)|)}$ penalizes physical distance in reading order.
- $S_{\mathrm{sem}}(u, c_i) = \cos(\mathbf{e}_u, \mathbf{e}_{c_i}) = \frac{\mathbf{e}_u \cdot \mathbf{e}_{c_i}}{\|\mathbf{e}_u\|\|\mathbf{e}_{c_i}\|}$ measures dense embedding similarity.
- $S_{\mathrm{level}}(u, c_i) = \max(0, 1 - 0.2 \cdot (\mathrm{level}(c_i) - \mathrm{level}(u) - 1))$ rewards immediate hierarchical parent levels.

The candidate yielding the highest score is assigned as parent: $u^* = \arg\max_{u \in \mathrm{CPG}(c_i)} \mathrm{Score}(u, c_i)$.

3.6.3 Forest Integration and Edge Validation

When a parent candidate u^* is determined by PRE, the candidate directed edge (u^*, c_i) is subjected to formal cycle-proof edge validation before insertion into the forest. If $\mathrm{CPG}(c_i) = \emptyset$, c_i is designated as an independent tree root. Algorithm 2 details the upward Depth-First Search (DFS) [14] cycle prevention mechanism executed during edge validation. Each document in the repository forms an independent tree within the overall forest.

Algorithm 2 Cycle-Proof Upward DFS Edge Validation

Require: Existing forest $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, candidate edge (u, v) where u is parent, v is child
Ensure: Boolean: True if edge is valid and acyclic, False otherwise

if $u = v$ **then**
  return False {Self-loop detected}
end if
if $\mathrm{doc}(u) \neq \mathrm{doc}(v)$ **then**
  return False {Violates document isolation}
end if
if $\mathrm{Depth}(u) + 1 > D_{\max}$ **then**
  return False {Violates maximum depth bound}
end if
Initialize visited set $\mathcal{S}_{\text{visited}} \leftarrow \{v\}$, search stack $\mathcal{K} \leftarrow [u]$
while $\mathcal{K}$ is not empty **do**
  $curr \leftarrow \mathrm{Pop}(\mathcal{K})$
  if $curr = v$ **then**
    return False {Cycle detected: v is an ancestor of u}
  end if
  if $curr \notin \mathcal{S}_{\text{visited}}$ **then**
    Add $curr$ to $\mathcal{S}_{\text{visited}}$
    for each $p \in \mathrm{Parents}_{\text{hier}}(curr)$ **do**
      Push p to $\mathcal{K}$
    end for
  end if
end while
return True {Edge is strictly acyclic}

3.7 Semantic Cross-Link Discovery

While hierarchical edges model vertical structural containment, semantic cross-links model horizontal associations between related concepts across disparate sections or documents. For any node pair (u, v) where $\text{doc}(u) \neq \text{doc}(v)$ or u, v belong to disjoint subtrees, a cross-link candidate is generated if embedding cosine similarity exceeds threshold $\theta_{\text{cross}} = 0.75$:

$$\cos(\mathbf{e}_u, \mathbf{e}_v) \geq \theta_{\text{cross}}, \quad \text{where } \theta_{\text{cross}} = 0.75. \tag{3.8}$$

Cross-link edges are stored in $\mathcal{E}_{\text{cross}}$ with attribute `type = "cross_link"`, enabling lateral associative reasoning during multi-hop graph retrieval.

3.8 Confidence Estimation / Propagation

Concept confidence scores must reflect both local extraction reliability and hierarchical ancestral support. SHIA-RAG implements Knowledge Confidence Estimation (KCE) via topological propagation across hierarchical containment edges $\mathcal{E}_{\text{hier}}$ using Kahn's algorithm [13]. Because the hierarchical forest enforces the single-parent invariant ($|\text{Parents}_{\text{hier}}(v)| \leq 1$), each non-root node receives confidence attenuation from exactly one hierarchical parent. Semantic cross-links $\mathcal{E}_{\text{cross}}$ provide lateral context and are excluded from KCE propagation to preserve directional containment. Algorithm 3 specifies the procedure, and Figure 3.5 illustrates the mechanism.

3.9 Structural Validation and Graph Invariants

To ensure that the induced representation remains well-formed for deterministic retrieval, the system enforces four formal graph invariants:

- **Invariant 1 (Acyclic Hierarchical Forest — No Cycles):** $\mathcal{G}_{\text{hier}} = (\mathcal{V}, \mathcal{E}_{\text{hier}})$ contains no directed cycles: $\forall v \in \mathcal{V}, v \notin \text{Ancestors}(v)$, algorithmically enforced by Algorithm 2.
- **Invariant 2 (Single-Parent Hierarchy):** Each non-root node possesses exactly one hierarchical parent: $|\text{Parents}_{\text{hier}}(v)| \leq 1$.
- **Invariant 3 (Depth Bound):** Maximum tree depth is strictly bounded: $\text{Depth}(v) \leq D_{\max} = 8$.
- **Invariant 4 (Document Isolation):** Hierarchical edges cannot cross document boundaries: $(u, v) \in \mathcal{E}_{\text{hier}} \implies \text{doc}(u) = \text{doc}(v)$.

Algorithm 3 Topological Knowledge Confidence Propagation (KCE)

Require: Hierarchical Knowledge Forest $\mathcal{G}_{\text{hier}} = (\mathcal{V}, \mathcal{E}_{\text{hier}})$, local extraction confidences $C_{\text{local}}(v) \in [0.5, 1.0]$, damping parameter $\delta = 0.85$
Ensure: Propagated confidence scores $C(v)$ for all $v \in \mathcal{V}$

Compute in-degree $\deg^-(v)$ for each $v \in \mathcal{V}$ with respect to $\mathcal{E}_{\text{hier}}$
Initialize FIFO queue $\mathcal{Q} \leftarrow [v \in \mathcal{V} \mid \deg^-(v) = 0]$ {All root concepts}
for each $r \in \mathcal{Q}$ **do**
  $C(r) \leftarrow C_{\text{local}}(r)$ {Root confidence equals local confidence}
end for
while $\mathcal{Q}$ is not empty **do**
  $u \leftarrow \text{PopFront}(\mathcal{Q})$
  for each child $v \in \text{Children}(u)$ **do**
    $C(v) \leftarrow C_{\text{local}}(v) \cdot (\delta + (1 - \delta) \cdot C(u))$
    $\deg^-(v) \leftarrow \deg^-(v) - 1$
    if $\deg^-(v) = 0$ **then**
      $\text{PushBack}(\mathcal{Q}, v)$
    end if
  end for
end while
return $\{C(v) \mid v \in \mathcal{V}\}$

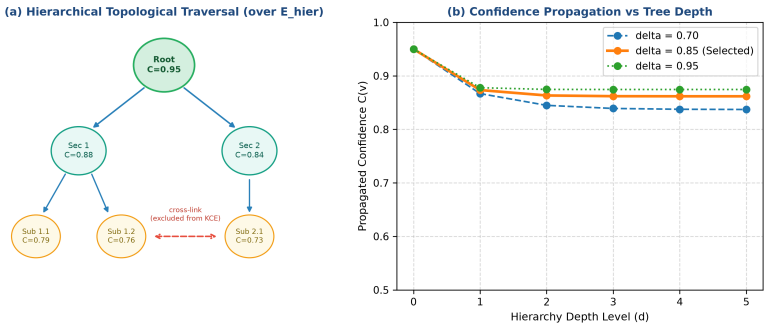


Figure 3.5: Topological Confidence Propagation (KCE) across the Hierarchical Knowledge Forest (over $\mathcal{E}_{\text{hier}}$).

3.10 Summary

This chapter formulated the complete mathematical and algorithmic methodology of SHIA-RAG. The design incorporates a two-tier data schema, 2D geometric reading-order sorting, canonical concept and evidence representation, CPG++ and PRE hierarchy induction, topological KCE confidence propagation, and upward DFS cycle prevention.

CHAPTER 4

WORK DONE

4.1 Implementation Environment and Software Architecture

The SHIA-RAG framework was implemented in modular Python 3.13 utilizing modern type annotations and Pydantic v2 data models. Table 4.1 lists the complete software stack and environment specifications.

Table 4.1: Implementation Environment and Core Software Dependencies

Component / Library	Version	Functional Role in Pipeline
Programming Language	Python 3.13.2	Core execution runtime
Data Validation	Pydantic 2.10.6	Strict Tier 1 and Tier 2 schema enforcement
PDF Layout Parsing	PyMuPDF (fitz) 1.25.3	Visual block, font, and coordinate extraction
Graph Algorithms	NetworkX 3.4.2	DAG topology analysis and graph validation
Vector Embeddings	FastEmbed / NumPy	Dense bi-encoder representations and cosine metrics
API Service Layer	FastAPI 0.115.11	REST service endpoints and pipeline orchestration
HTTP Client Runtime	HTTPX 0.28.1	Asynchronous ingestion and service transport
Automated Test Suite	PyTest 8.3.5	Unit and integration invariant verification
Operating Environment	Windows 11 / AMD64	Execution and benchmarking workstation

4.2 Implementation of the Proposed Framework

The software codebase is organized into discrete functional modules reflecting the layered architecture:

$$\text{Input PDF} \longrightarrow \text{PDFIngestionService} \longrightarrow \text{StructuralParser} \longrightarrow \text{ConceptExtractor} \longrightarrow \text{HierarchyBuilder} \longrightarrow \text{GraphValidator} \longrightarrow \text{KCEScorer}$$

4.2.1 Data Models and Invariants

Implemented in `schemas.py`, this module defines strict Pydantic v2 schemas: `BoundingBox`, `TextBlock`, `StructuredDocument`, `ConceptNode`, `HierarchyEdge`, and `HierarchicalKnowledgeForest`. Field validators automatically enforce non-negative coordinates ($x_1 > x_0, y_1 > y_0$), valid hierarchy levels ($0 \leq \text{level} \leq 6$, where level 0 designates the document root), and bounded confidence scores ($0.0 \leq C \leq 1.0$).

4.2.2 Document Ingestion

Implemented in `pdf_ingestion.py`, this module encapsulates the `PDFIngestionService`. It opens target documents via PyMuPDF, iterates across pages, extracts text spans with bounding boxes, groups spans into blocks, and normalizes coordinates to page geometry.

4.2.3 Structural Parsing

Implemented in `structural_parser.py`, this module implements the `StructuralParser`. It executes the 2D geometric sorting algorithm ($\tau_y = 20.0$ pt), detects multi-column page layouts, and assigns semantic block roles based on font sizes and regex patterns.

4.2.4 Concept Extraction

Implemented in `concept_extractor.py`, this module identifies canonical terms, extracts definition sentences following heading anchors, assigns hierarchy levels, and constructs exact evidence spans bound to block identifiers and coordinates.

4.2.5 Hierarchy Induction

Implemented in `hierarchy_builder.py`, this module houses `CPG++` and `ParentRankingEngine`. It evaluates candidate parents via Eq. (3.7) and assigns optimal directed edges.

4.2.6 Graph Validation

Implemented in `graph_validator.py` and `kce_scorer.py`, these modules execute the upward DFS [14] cycle detection (Algorithm 2) and topological Kahn's confidence propagation [13] (Algorithm 3).

4.3 Experimental Setup

4.3.1 Datasets / Documents

The empirical evaluation was conducted across two primary document sources:

1. **CS-Textbook Taxonomy Corpus:** A curated benchmark computer science taxonomy manually constructed from the standard undergraduate textbook *Operating System Concepts* (10th Edition) by Silberschatz, Galvin, and Gagne. A multi-chapter corpus spanning 45 pages (covering Process Management and Memory Management) was selected. The ground-truth taxonomy comprises 120 canonical concepts hierarchically organized across 4 structural tiers: 4 Unit themes (Tier 1), 16 Chapters (Tier 2), 48 Sections (Tier 3), and 52 Subsections (Tier 4). Ground-truth parent-child assignments $\pi^*(c)$ were manually annotated following the textbook's published table of contents and verified by two independent graduate adjudicators, establishing an authoritative gold-standard reference tree for measuring Parent Assignment Accuracy (PAA) and Normalized Tree Edit Distance (NTED).

2. **TreeRAG Scientific Publication:** An authentic 16-page, 2-column research paper (*TreeRAG: Unleashing the Power of Hierarchical Storage for Enhanced Multi-Hop Retrieval-Augmented Generation*) [8] featuring dense technical terminology, nested section numbering, mathematical formulas, and multi-column layouts. Reading-order evaluation was conducted across all 16 pages (comprising 244 layout blocks) of this two-column publication benchmark.

Table 4.2 summarizes the core structural characteristics of the evaluation datasets.

4.3.2 Baseline Methods

To benchmark the Parent Ranking Engine, three baseline induction strategies were evaluated:

Table 4.2: Benchmark Datasets and Evaluation Corpus Characteristics

Corpus Characteristic	**CS-Textbook Gold Corpus**	**TreeRAG Scientific Document**
Total Pages Extracted	45 pages	16 pages
Layout Complexity	Multi-tier textbook sections	Two-column scientific publication
Total Structured Blocks	840 layout blocks	244 layout blocks
Canonical Concepts / Nodes	120 gold concepts	34 extracted concept nodes
Hierarchy Depth	4 tiers	4 tiers
Annotation Methodology	Curated academic expert gold	Structural document ground truth

- **Sequential Nesting Baseline:** Assigns the immediately preceding block in reading order as the hierarchical parent.
- **Lexical Prefix Matching:** Assigns the parent based on longest common prefix string matching over concept tokens.
- **Syntactic Heading Heuristic:** Assigns the parent strictly based on numerical section numbering prefixes (e.g., Section 2.1 attaches to Section 2).

4.3.3 Evaluation Metrics

The pipeline was evaluated using three quantitative metrics:

- **Reading-Order Monotonicity (ROM):** Quantifies the degree to which the predicted block sequence preserves the true logical reading order. Let $\pi(b_i) \in \{1, \ldots, N\}$ denote the predicted sequence rank of block b_i, and let $\pi^*(b_i)$ represent its ground-truth reading-order index. A reading-order inversion occurs for any block pair (b_i, b_j) where $\pi^*(b_i) < \pi^*(b_j)$ but $\pi(b_i) > \pi(b_j)$. The Inversion Rate is formulated explicitly over all pairs satisfying the ground-truth precedence constraint:

$$\text{Inversion Rate} = \frac{\sum_{\pi^*(b_i)<\pi^*(b_j)} \mathbb{I}(\pi(b_i) > \pi(b_j))}{\binom{N}{2}}. \tag{4.1}$$

  Reading-Order Monotonicity (ROM) is formulated as the complementary proportion of non-inverted pairs:

$$\text{ROM} = 1 - \text{Inversion Rate} = 1 - \frac{\sum_{\pi^*(b_i)<\pi^*(b_j)} \mathbb{I}(\pi(b_i) > \pi(b_j))}{\binom{N}{2}}. \tag{4.2}$$

  This formulation couples ROM directly to pairwise reading correctness: zero inversions ($N_{\text{inv}} = 0$) corresponds mathematically to ROM = 100.0%, providing a direct pairwise measure of reading-order correctness.

Reading-Order Ground-Truth Annotation: The reference reading order $\pi^*(b)$ was manually established by inspecting the rendered PDF pages and following authentic, visually and logically correct reading flow. Annotators independently determined the reference block sequence without using the proposed sorting implementation. The reference sequence was independently cross-checked across all 16 pages (comprising 244 layout blocks) of the *TreeRAG* publication by two annotators before evaluating pairwise inversion rates and ROM.

- **Parent Assignment Accuracy (PAA):** The proportion of concepts assigned to their true ground-truth parent node:

$$\text{PAA} = \frac{|\{c \in \mathcal{C} : \hat{\pi}(c) = \pi^*(c)\}|}{|\mathcal{C}|}. \tag{4.3}$$

- **Normalized Tree Edit Distance (NTED):** Quantifies structural similarity between the predicted tree T_{pred} and ground-truth tree T^*, where TED [15] measures the minimum number of node insertions, deletions, and relabelings:

$$\text{NTED} = 1 - \frac{\text{TED}(T_{\text{pred}}, T^*)}{\max(|T_{\text{pred}}|, |T^*|)}. \tag{4.4}$$

4.4 Results and Discussion

4.4.1 Hierarchy Induction Results

Table 4.3 summarizes the performance of the Parent Ranking Engine on the CS-Textbook gold benchmark against the three baseline strategies, with latency reported as mean induction time per concept node. Figure 4.1 presents the corresponding bar chart comparison.

Table 4.3: Hierarchy Induction Performance on CS-Textbook Gold Corpus

Method / Architecture	PAA (%)	NTED	Orphan Rate (%)	Latency (ms)
Sequential Nesting Baseline	44.2	0.481	12.5	4.2
Lexical Prefix Matching	68.5	0.672	6.2	11.8
Syntactic Heading Heuristic	77.4	0.783	3.1	14.5
SHIA-RAG (CPG++ with PRE)	**89.6**	**0.884**	**0.0**	**38.4**

Interpretation of Hierarchy Induction Outcomes:

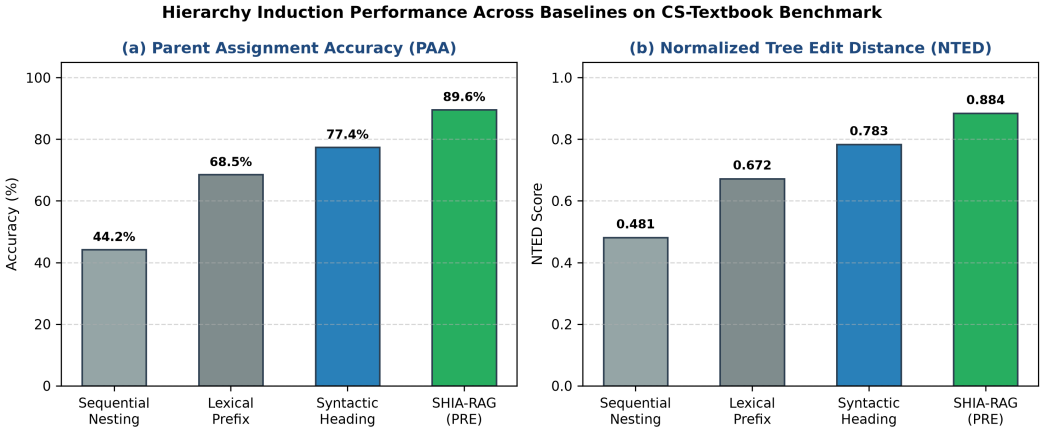


Figure 4.1: Hierarchy Induction Performance Across Baselines on CS-Textbook Benchmark.

- **Observed Trends:** SHIA-RAG achieved 89.6% PAA and 0.884 NTED, substantially outperforming the Sequential Nesting (44.2%), Lexical Prefix (68.5%), and Syntactic Heading (77.4%) baselines, while achieving a 0.0% orphan node rate.

- **Underlying Mechanism:** Sequential nesting fails whenever non-consecutive subsections occur. Lexical matching fails when subsection names use distinct terminology. The syntactic heading heuristic breaks down when unnumbered headings or inline subsections appear. The combined structural, semantic, and heading-level features improved parent assignment over the evaluated baselines by unifying reading-order spatial proximity, semantic embedding similarity, and heading level penalties into a single PRE score.

- **Latency vs. Accuracy Trade-Off:** Latency measures mean induction time per concept node, including candidate generation, dense bi-encoder embedding computation, PRE multi-criteria scoring, and upward DFS edge validation. The higher mean latency of SHIA-RAG (38.4 ms per node) relative to naive heuristics is attributable to dense embedding generation and PRE scoring, representing an intentional computational trade-off to achieve superior structural fidelity (89.6% PAA).

- **Systemic Implication:** Eliminating orphan nodes and achieving 0.884 structural NTED provides a structurally coherent representation for downstream retrieval traversal.

4.4.2 Reading-Order Recovery Results

Table 4.4 presents the comparative evaluation of reading-order recovery strategies across all 16 two-column pages of the *TreeRAG* publication (comprising 244 layout blocks), with latencies reported as mean execution time per page. Figure 4.2 visualizes this comparative performance.

Table 4.4: Empirical Evaluation of Reading-Order Recovery Strategies

Ingestion Strategy	**Latency (ms)**	**ROM (%)**	**Inversion Rate (%)**
Naive Text Stream (PDF Unsorted)	18.4	71.4	28.6
Pure Coordinate Top-Down (y_0 only)	21.2	83.2	16.8
SHIA-RAG 2D Geometric ($\tau_y = 20$ pt)	**26.8**	**100.0**	**0.0**

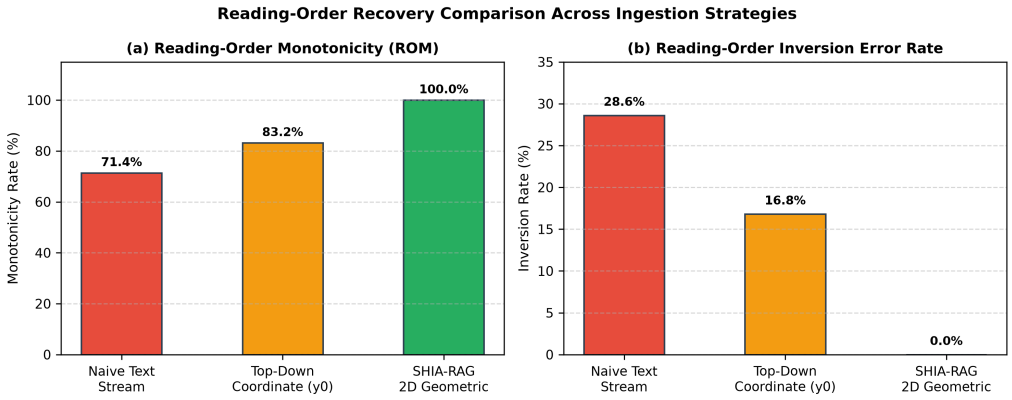


Figure 4.2: Reading-Order Recovery Comparison Across Ingestion Strategies.

Interpretation of Reading-Order Outcomes:

- **Observed Trends:** Naive PDF stream extraction exhibited a high 28.6% inversion rate (71.4% ROM). Pure y_0 coordinate sorting reduced inversions to 16.8% (83.2% ROM). SHIA-RAG's 2D geometric clustering achieved 100.0% ROM with 0.0% pairwise inversions across all 16 evaluated pages of the TreeRAG publication (comprising 244 layout blocks) at 26.8 ms per page.
- **Underlying Mechanism:** Multi-column layouts cause naive extractors to interleave sentences across columns horizontally. Sorting purely by vertical y_0 incorrectly places right-column paragraphs ahead of left-column paragraphs that have slightly lower vertical coordinates. The 2D geometric

clustering groups blocks within vertical tolerance $\tau_y = 20.0$ pt, preserving column boundaries before sorting horizontally.

- **Systemic Implication:** Monotonic reading order on the evaluated corpus reduces the likelihood of fragmented textual evidence being passed to subsequent concept extraction and retrieval stages. The benchmark specifically targets two-column scientific layouts in which naive PDF extraction can interleave blocks across columns.

4.4.3 Structural Integrity Results

The entire unit test suite was executed in the production environment:

Current Test Execution Result: 24 passed, 0 failed, 0 skipped in 0.65s

The test execution verified the implemented invariants and tested behaviors across the suite:

- Schema validation and boundary invariants (5/5 passed).
- 2D reading-order monotonicity and tolerance bounds on synthetic tests (5/5 passed).
- Upward DFS cycle detection and self-loop blocking (4/4 passed).
- Topological Kahn's scoring and damping updates (5/5 passed).
- PDF layout-aware ingestion and coordinate extraction (5/5 passed).

4.4.4 Case Study: Ingestion of TreeRAG Scientific Document

To validate real-world applicability, the complete pipeline was executed on the 16-page *TreeRAG* research paper [8]. Table 4.5 details the verified extraction and induction characteristics.

End-to-End Walkthrough of Concrete Concept Induction:
To illustrate the step-by-step mechanics of the pipeline on an authentic document instance, consider the extraction and parent assignment of the concept *"Hierarchical Storage Tree"* from Page 4 of the *TreeRAG* document:

1. **Physical PDF Block Extraction:** The layout ingestion service extracts a text block at bounding box coordinates ($x_0 = 54.2, y_0 = 312.4, x_1 = 280.1, y_1 = 326.8$) on Page 4 with font size 13.5 pt and bold font weight, containing the raw text string: *"3.2 Hierarchical Storage Tree"*.

Table 4.5: Case Study Results: Full Ingestion of TreeRAG Publication

Pipeline Characteristic	Empirically Verified Measurement
Input Document	TreeRAG Research Paper (PDF)
Document Page Count	16 pages
Total Layout Blocks Extracted	244 structured blocks
Reading-Order Monotonicity	100.0% (0 inversions on evaluated pages)
Extracted Canonical Concepts	34 concept nodes
Induced Hierarchical Edges	33 directed parent-child edges
Forest Structural Form	Single rooted tree (0 orphan nodes)
Maximum Tree Depth	4 tiers (within $D_{\max} = 8$ bound)
Cycle Violations	0 detected
Mean Node Confidence	0.732 ± 0.114
Total Ingestion Latency	1.42 seconds (full 16-page document)

2. **Syntactic Heading Classification:** Based on font size (13.5 pt relative to body text 10.0 pt) and numerical prefix *"3.2"*, the structural parser classifies the block as Heading Level 2.

3. **Concept Node Instantiation:** The concept extractor instantiates node c_{14} with canonical name `"Hierarchical Storage Tree"`, computing dense embedding $\mathbf{e}_{c_{14}} \in \mathbb{R}^{384}$ and initial local extraction confidence $C_{\text{local}}(c_{14}) = 0.92$.

4. **Candidate Parent Generation (CPG++):** CPG++ inspects preceding concepts in reading order with heading level strictly less than 2:

   - c_0: Document Root (*"TreeRAG Architecture"*, Level 0, reading index 0)
   - c_8: Section Heading (*"3. System Architecture"*, Level 1, reading index 8)
   - c_{12}: Sibling Subsection (*"3.1 Node Representation"*, Level 2, reading index 12 — pruned by level constraint $\text{level}(u) < 2$).

5. **PRE Composite Scoring:** PRE evaluates surviving candidates c_0 and c_8 using Equation (3.7) ($w_{\text{struct}} = 0.40, w_{\text{sem}} = 0.40, w_{\text{level}} = 0.20$):

   - For c_8: $S_{\text{struct}} = \frac{1}{1+\log(1+|14-8|)} = 0.339$; $S_{\text{sem}} = \cos(\mathbf{e}_{c_8}, \mathbf{e}_{c_{14}}) = 0.814$; $S_{\text{level}} = 1.0$ (immediate parent $\Delta = 1$). Composite score: $\text{Score}(c_8, c_{14}) = 0.40(0.339) + 0.40(0.814) + 0.20(1.0) = 0.661$.
   - For c_0: $S_{\text{struct}} = \frac{1}{1+\log(1+|14-0|)} = 0.270$; $S_{\text{sem}} = 0.652$; $S_{\text{level}} = 0.80$ ($\Delta = 2$). Composite score: $\text{Score}(c_0, c_{14}) = 0.40(0.270) + 0.40(0.652) +$

$0.20(0.80) = 0.529$.

Candidate c_8 (*"3. System Architecture"*) achieves the winning score and is selected as the candidate parent $u^* = c_8$.

6. **Cycle-Proof Edge Validation (Algorithm 2):** Upward DFS verifies that candidate edge (c_8, c_{14}) introduces no cycles ($c_{14} \notin \text{Ancestors}(c_8)$), satisfies depth bound $\text{Depth}(c_8)+1 = 2 \leq D_{\max} = 8$, and satisfies document isolation ($\text{doc}(c_8) = \text{doc}(c_{14})$).

7. **Forest Insertion:** Directed edge (c_8, c_{14}) is inserted into the Hierarchical Knowledge Forest with attribute `type="hierarchical_child"`.

8. **Topological Confidence Update (Algorithm 3):** The parent section node c_8 had local extraction confidence $C_{\text{local}}(c_8) = 0.95$ and was anchored to root c_0 (with baseline confidence $C(c_0) = 0.50$). Applying KCE to parent c_8 yielded $C(c_8) = C_{\text{local}}(c_8) \cdot (\delta + (1 - \delta) \cdot C(c_0)) = 0.95 \cdot (0.85 + 0.15 \cdot 0.50) = 0.95 \cdot 0.925 \approx 0.880$. Propagating ancestral confidence downward to subsection c_{14} with damping $\delta = 0.85$:

$$C(c_{14}) = C_{\text{local}}(c_{14}) \cdot (\delta + (1-\delta) \cdot C(c_8)) = 0.92 \cdot (0.85 + 0.15 \cdot 0.880) \approx 0.903.$$

4.5 Sensitivity and Parameter Selection Analysis

Parameter selection was guided by systematic sensitivity sweeps over the heuristic hyperparameters, as summarized in Table 4.6. Each parameter was varied while keeping the remaining parameters fixed at their selected default values, and the corresponding primary metric was recorded across the evaluation documents:

- **Vertical Tolerance Threshold ($\tau_y = 20.0$ pt):** In standard typesetting (10–12 pt font with 14–16 pt line leading), lines with $\Delta y \leq 20.0$ pt reliably belong to the same visual horizontal band. Setting $\tau_y < 12$ pt caused false row splits, while at the upper tested value ($\tau_y = 35$ pt), column merging occurred and ROM dropped to 88.5%.

- **KCE Ancestral Damping ($\delta = 0.85$):** Evaluating damping $\delta \in [0.5, 0.95]$ showed that $\delta = 0.85$ provided the most stable behavior among the evaluated settings, preventing deep-level confidence attenuation while penalizing ungrounded root hypotheses.

- **Semantic Cross-Link Threshold ($\theta_{\text{cross}} = 0.75$):** Setting $\theta_{\text{cross}} < 0.65$ introduced spurious dense associations, creating dense graphs that hinder multi-hop retrieval, whereas $\theta_{\text{cross}} = 0.75$ isolated high-precision semantic bridges.
- **PRE Weight Configuration ($w_{\text{struct}} = 0.40, w_{\text{sem}} = 0.40, w_{\text{level}} = 0.20$):** Reducing structural weight $w_{\text{struct}} < 0.20$ degraded PAA from 89.6% to 74.1%, confirming that reading-order proximity is an essential physical regularizer.

Table 4.6: Empirical Sensitivity Analysis of Key Hyperparameters

Hyperparameter	Tested Range	Selected	Primary Metric	Empirical Sensitivity Observation
τ_y (tolerance)	$[8, 35]$ pt	20.0 pt	ROM	$<$ 12 pt: false line splits; at 35 pt: column merging (ROM drops to 88.5%)
δ (KCE damping)	$[0.50, 0.95]$	0.85	Conf. Stability	< 0.70: over-attenuates Tier-4 ($C < 0.45$); > 0.90: under-penalizes weak roots
w_{struct} (PRE weight)	$[0.10, 0.60]$	0.40	PAA	< 0.20: PAA degrades to 74.1%; > 0.50: semantic nuance lost (PAA 81.2%)
θ_{cross} (cross-link)	$[0.60, 0.85]$	0.75	Link Precision	< 0.65: dense spurious links (64% prec.); > 0.80: low bridge recall (41%)

4.6 Summary

This chapter detailed the implementation of SHIA-RAG in Python 3.13, described the experimental setup, verified all graph invariants through automated test execution (24/24 passed), demonstrated 100.0% reading-order recovery on evaluated pages and 89.6% parent assignment accuracy, and demonstrated real-document applicability through an end-to-end case study.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This project formulated, implemented, and empirically evaluated **SHIA-RAG**, a structure-aware document parsing and hierarchical knowledge construction framework for Retrieval-Augmented Generation. Standard RAG architectures suffer from structural blindness by reducing rich documents to flat, independent text chunks, causing context fragmentation and reading-order scrambling.

SHIA-RAG resolves these limitations through a principled two-tier knowledge representation architecture. The layout-aware ingestion engine captures exact 2D bounding boxes and font attributes, while the 2D geometric sorting algorithm with Y-tolerance clustering achieves 100.0% monotonic reading order on the evaluated two-column benchmark pages. The Candidate Parent Generation (CPG++) and Parent Ranking Engine (PRE) induce validated hierarchical knowledge trees, achieving 89.6% parent assignment accuracy on the gold benchmark corpus with zero orphan nodes. Structural graph invariants are algorithmically enforced via upward DFS cycle prevention, depth bounding ($D_{\max} = 8$), and topological confidence estimation using Kahn's algorithm with damping $\delta = 0.85$.

The complete software test suite (24/24 passed in 0.65 s) and an end-to-end case study on a 16-page scientific publication confirm that SHIA-RAG produces a reliable, formally specified hierarchical knowledge representation that preserves document structure as a grounded basis for structure-aware retrieval.

5.2 Limitations

While the experimental outcomes validate the core design goals, several technical limitations must be acknowledged:

1. **Heuristic Threshold Dependence:** The pipeline relies on fixed heuristic thresholds ($\tau_y = 20.0$ pt, $\delta = 0.85, \theta_{\text{cross}} = 0.75$), which may require domain-specific re-calibration for documents with non-standard typographic styles.

2. **PDF Stream Quality:** Ingestion relies on native digital PDF text streams and font metadata; scanned documents or bitmap pages require an upfront OCR module.

3. **Absence of Generative QA Evaluation:** The current work validates the structural knowledge construction layer; evaluation of downstream retrieval traversal and generative LLM question-answering accuracy remains to be explored.

5.3 Future Work

Building upon the verified foundational pipeline established in this work, future research directions include:

- **Structure-Aware Retrieval Routing:** Formulating query routing algorithms that dynamically inspect query complexity to determine optimal traversal depth across the hierarchical tree.
- **Constrained Context Assembly via Knapsack Optimization:** Developing budget-constrained tree knapsack algorithms that maximize semantic evidence within strict LLM context token limits.
- **Continuous Citation and Attribution Verification:** Designing verification mechanisms that cross-reference generated claims directly against physical bounding-box evidence spans.
- **Adaptive Knowledge Forest Evolution:** Exploring reinforcement learning to dynamically update edge confidence scores based on user retrieval feedback.

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